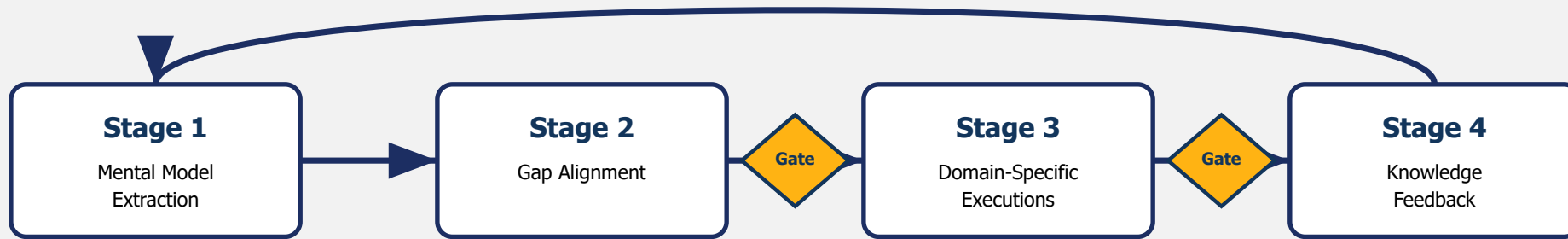


Skill-Based Agent Automation Methodology

- Big Picture
- Stage 1: Mental Model Extraction
- Stage 2: Gap Alignment
- Stage 3: Domain-Specific Execution
- Gate: Human Review and Decision
- Stage 4: Knowledge Feedback

Big Picture



- **Stage 1 (Mental Model Extraction):**

- Scattered specs, codes → unified shared, queryable understanding.

- **Stage 2 (Gap Alignment):**

- Manual gap-spotting → scoped, actionable gaplist automatically

- **Stage 3 (Domain-Specific Execution):**

- Split works by domain expertises.

- **Gate (Human Review):**

- Micromanaging steps → human decisions at points matter most.

- **Stage 4 (Knowledge Feedback):**

- From scratch → continuously improving workflow efficiency.

Stage 1: Mental Model Extraction

Agent Conversion (Mental Model Extraction Skill)

- Input
 - Existing filelists, top module, specifications.
- Output
 - Semantic Markdown Cachings.
 - Protocols, hardware hierarchy, FSM control, registers.
 - Linked to structured AST nodes.

Mental Model

- Semantic markdown cachings.
- Structured AST/.db database.
- Database.

Type1 Conversion: Structured Data (Parsing Skill)

- Input
 - Exiting code / waveform.
- Output
 - Parsed structured AST / .db database.

Type2 Conversion: Database (Knowledge Indexing Skill)

- Input (Empty Initially)
 - Best practice, history failure, documents from Stage 4.
- Output (Small)
 - Flatten markdown files.
- Output (Large)
 - RAG-liked Files.
 - Chunking → indexing → vector embedding
 - Note: open-source Apache 2.0 licensed, 30k+ GitHub stars.

Stage 2: Gap Alignment

Artifacts Derivation (Derivation Skill)

- Input
 - Mental models from Stage 1.
- Output
 - Mental-model-derived artifacts.
 - Mental-model-derived testplan, ...

Alignment (Alignment Skill)

- Input
 - Existing artifacts.
 - Specifications, regression testplans, coverages.
- Output
 - Verified linkage of alignment among existing artifacts.

Without stage 2, human alignment is mandatory

Gap Analysis (Gap Analysis Skill)

- Input
 - Tasks prompt.
 - Mental model from Stage 1.
 - Mental-model-derived artifacts.
 - Existing artifacts.
 - Verified linkage of alignment among existing artifacts.
- Output
 - Scoped, Actionable GapList.
 - Regression testplan vs. mental-model-derived testplan.
 - **Human Decision:** chase, waive, skip

Stage 3: Domain-Specific Execution

Domain-Specific Update (Update Skills)

- Input
 - Scoped, actionable gaplist from Stage 2.
 - Existing artifacts.
 - Mental model from Stage 1.
- Output
 - Updated artifacts via matched domain skill in scope of gaplist.
 - **CDC, Power, DFT, MBIST, Replay, Golden-Sim Skill.**

Domain-Specific Execution (Execution Skills)

- Input
 - Updated artifacts.
- **Human Review:** Output
 - Task success: targeted gap resolved
 - Regression safety: nothing else broken

Workflow (Skill-Defined Procedure)

- Step-by-step procedure written in skill
- Run tool and script → read report → check rules → summary

Gate: Humans Review and Decision

Well-Defined Human Review Key Points

- Stage-by-Stage Maturity
 - Evidence-based progression
 - Prove one stage before advancing
- Gate 1 (Stage 2 → 3):
 - Reviews
 - Gaplist, priority, confidence, cost/effort
 - Decides
 - To chase, to waive or to skip
- Gate 2 (Stage 3 → 4):
 - Reviews
 - Diff, scope, task success, regression safety
 - Decides
 - Commit Authorization
 - Loop back to Stage 3 if not authorized
 - Full Case Knowledge Package
 - Representative and generalizable to future gaps

Stage 4: Knowledge Feedback

Knowledge Storage (Knowledge-Feedback Skill)

- Input
 - Full Case Knowledge Package
 - Gap ID, fix, evidence, round metadata
- Output
 - Pattern Library & Knowledge Base
 - Accumulated Record Chain
 - Gap ID, root cause, fix
 - Reusable Debug History
 - Traceable model, loop, change, commit hash

Scale-Appropriate Database

- Small-Scale Setup
 - Flatten markdown file, keyword search
- Large-Scale Setup
 - RAG-liked: chunking, embedding, vector database

Key link to close the loop

System never gets better without it.